

Model Card - ICU Mortality Prediction (Random Forest)

Version: 1.0 | Date: 2025-12-31 | Intended use: portfolio demonstration (not for clinical deployment)

Summary

This project builds an explainable, calibrated ICU mortality prediction model using a public ICU time-series dataset. It demonstrates end-to-end ML delivery: data parsing, feature engineering, missingness handling, model training, calibration, threshold selection for a clinical operating point, and interpretability using local feature attributions.

Important: This model is a learning and portfolio artifact. It is not validated for clinical use.

1. Data

Source: Public ICU dataset (PhysioNet/Kaggle-style format) consisting of per-patient text files with irregularly sampled measurements across the first 48 hours and a separate outcomes file mapping patient_id to mortality label.

Cohort size: 4,000 patients

Class balance: 3,446 survivors (0), 554 deaths (1) - mortality prevalence ~13.9%

2. Features and preprocessing

Raw time-series were converted into a tabular feature set (one row per patient). For each clinical variable, the pipeline derived mean, last, and was_measured indicators. Special handling was applied to selected variables (e.g., FiO2 scaling and MechVent proportion). Missingness was retained and then imputed (median for numeric features; 0 for measurement flags).

Final feature table: 63 columns (including patient_id and outcome); 61 model features.

3. Model

Algorithm: RandomForestClassifier

Key settings: n_estimators=300, max_depth=None, class_weight='balanced', random_state=42

Split strategy: stratified 80/20 split by patient (Train 3,200; Test 800)

4. Performance

Metric	Result
Test ROC-AUC	0.844
5-fold CV ROC-AUC (mean ± SD)	0.823 ± 0.018
PR-AUC	0.461

5. Calibration and operating threshold

Isotonic calibration (5-fold) was applied. The operating threshold was selected to target approximately 60% recall for deaths (sensitivity). Chosen threshold: 0.223.

At threshold 0.223 (calibrated)	Predicted survival	Predicted death
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True survival (0)	595	94
True death (1)	44	67

At this operating point: precision (death) = 0.416, recall (death) = 0.604, overall accuracy = 0.828. This reflects a deliberate safety-oriented trade-off: higher sensitivity at the cost of more false positives.

6. Interpretability (local examples)

Local feature attributions were reviewed for representative cases. Two examples (one correctly flagged death, one missed death) showed clinically plausible drivers of risk.

Example A - death correctly flagged (high-risk physiology)

Feature	SHAP contribution	Feature value
GCS_last	+0.0568	8
BUN_last	+0.0462	104
Lactate_last	+0.0397	4.9
HCO3_last	+0.0279	15
NISysABP_last	+0.0178	86

Interpretation: low GCS, high BUN, raised lactate, and low bicarbonate pushed predictions towards death, consistent with organ dysfunction.

Example B - death missed (false negative)

Feature	SHAP contribution	Feature value
GCS_last	+0.0704	8
BUN_mean	-0.0354	18.75
Lactate_last	-0.0268	1.5
Creatinine_mean	-0.0217	0.9
pH_last	-0.0139	7.42

Interpretation: low GCS increased risk, but reassuring metabolic/renal markers reduced predicted risk, illustrating limitations of summary features.

7. Limitations and intended use

- Dataset-specific; results depend on cohort definition and measurement patterns.
- Mean/last summarization discards fine-grained trajectories.
- Mortality is heterogeneous, limiting predictability from early physiology alone.
- Portfolio artifact only; clinical deployment would require prospective evaluation, workflow integration, safety case, and governance approvals.